There is plenty of information around - but often too much of the wrong kind and not enough if the right kind.

MARKETING INFORMATION SYSTEMS

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The Nature of Marketing

Marketing management is about finding ways of satisfying customer wants and needs, while achieving organizational objectives or requirements in terms of profit or some other measure of corporate performance. It brings together all customer-impinging resources, such as product design and specification, advertising and other forms of promotion, pricing policy, selling, channels of distribution and physical distribution to achieve this end. These customer-impinging resources are often summarized under four headings and referred to as the four Ps of the marketing mix (price, promotion, place and product). The art or science of marketing management is concerned with making decisions/ policies with respect to the elements of the' marketing mix such that the company's interface with its markets is both profitable and customer satisfying.

The Need for Marketing Information

managers require information to help them forecast changes in product demand, increase selling Productivity, and exercise control over sales and distribution expenses. Marketing is an ongoing process; decisions are made and results of these decisions have he monitored. Consumer and competitor reactions to the company's decisions have to be studied to ensure that the best strategy is being employed. Information on these matters is used to correct deviations from plans. For example, if a target of a 5 per cent increase in new clients has been set for the salesforce then it is necessary to monitor how effective the salesforce is in terms of reaching the target set. Such information can then be used to adjust targets, if the need arises.

Information is needed for decision making. Unfortunately, in many firms, it is often difficult to obtain information of the right kind. The kinds of complaints one often encounters are [1]:

- There is too much information of the wrong kind.
- There is not enough information of the right kind.
- Information is too dispersed to be useful.
- Information arrives too late to be useful.
- Information often arrives in a form that leaves no idea of its accuracy and therefore lacks credibility.

Clearly, there is a need to overcome these kinds of problems and complaints and it is for this reason that marketing information systems have evolved.

Marketing Information Systems

The concept of marketing information systems has been around for many years. Early systems were paper-based systems but, with the emergence of computers with large storage capacities and later microcomputers with similar features, marketing information systems have become more "electronic" in nature. MIS (marketing information systems) can be classified under five headings:

- Planning systems - which provide information on sales, costs and competitive activity, together with any kind of information which is needed to formulate plans.
- Control systems - these provide continuous monitoring of marketing activities and enable marketing executives to identify problems and opportunities in the marketplace. At the same time, they permit a more detailed and comprehensive review of performance against plans.
- Marketing research systems - such systems allow executives to test decision rules and cause/effect hypotheses. This permits the assessment of the effects of marketing actions and encourages improved learning from experience.
- Monitoring systems - these systems provide management with information concerning the external environment in which they are operating.

One can define a marketing information system as one which scans and collects data from the environment, makes use of data from transactions and operations within the firm and then filters, organizes and selects data before presenting them as information to management.

Marketing Information Systems in the United Kingdom

Many firms in the UK are starting to develop computer-based marketing information systems. A survey by Martech Ltd [2] of UK firms found that, in the companies they surveyed, computerization had grown very rapidly over the previous two years with penetration increasing from 15 per cent to about 50 per cent - that is, 70 per cent of the systems mentioned in the survey had been implemented in the previous two years. Martech also noted that most of the growth had been concentrated on fragmented solutions to tackle a particular element of the marketing/sales function with a particular emphasis on salesforce productivity. Few companies had moved beyond the isolated productivity tool stage towards integrated marketing information systems covering customer service, account management, product management, forecasting, sales management, advertising and promotion, distribution, pricing, competitive tracking or marketing research.

Recently, the writer contacted 20 medium- to large-sized firms, based north of Birmingham, to enquire as to the state of their marketing information systems. Only six firms responded to the request for information and a visit was arranged to three of them to discuss the nature of their marketing information systems.

The first of the six firms indicated that a system has currently being designed by the marketing department and that it would be capable of handling the following tasks:

- Setting prices and evaluating different pricing strategies.
- Monitoring and evaluating new product-market opportunities.
- Evaluating the optimality of the current product-market portfolio.
- Analyzing customer accounts.
- Planning, analyzing and evaluating sales activities.
- Market measurement and sales forecasting.
- Quantitative aspects of market research.

The firm indicated that the system had not currently been designed to plan, monitor or control the promotional activities of the firm, although it indicated that this might be included at a later date. My subsequent visit to the firm proved rather fruitless since, although it was working on the development of a system, it was not keen to discuss progress in any detail.

Firm number two was developing a marketing information system too. I saw little evidence of it when I visited the firm.

Firm number three indicated that it had a marketing information system in operation. When I visited the firm I found there was some use of decision support systems. I found a microcomputer being used for project management planning. The package in use was being applied specifically in connection with new development and made use of PERT analysis. At the time of the visit, the people concerned were still learning to use the package. There was also a relatively sophisticated forecasting package in use. This was again microcomputer based and was employed to forecast demand for existing product lines. There was also a "home-made" microcomputer-based database system, written in BASIC, in which was kept all the data on product prices, discounts, customer records etc. The company also had a mainframe-based system which was connected remotely to a PC in each one of a number of distributor outlets. Every night the system was able to scan the records held on the PCs and the following morning produce a report for top management on the sales of the previous day at the distributor outlets. Unfortunately, the system was only partially complete at the time of my visit.

COMPUTERIZED MIS IN THE UK ARE STILL IN THE EARLY STAGES OF DEVELOPMENT

I was unable to visit the three other firms but they did provide me with data on the state of their marketing information systems. Firm number four indicated that it was using microcomputers in a number of different art' s of marketing activity. The firm was in the retailing business and had modelled store sales so as to be able to estimate new site potential. It was also using lap-top computers in stores for survey interviews with customers.

Firm number five was actively into database marketing. Its database was used for:

- Analyzing customer accounts: customers were graded by size and credit-worthiness and recorded geographically in the database. SIC code and employee size were also shown.

Customers were analyzed by product use and this was used to target them with relevant mailshots.

- Planning, analyzing and evaluating sales activities: leads that were generated through direct marketing activity were allocated by computer to the appropriate salesman. The result of each enquiry was monitored and analyzed by product and sales area. This, the firm felt, enabled it to evaluate the success of its salesforce and that of campaigns.
- Promotional activities: all enquiries/leads generated by advertising, PR, direct mail and telemarketing were analyzed to determine revenue, cost per sale, and cost per enquiry.

Firm number six indicated that, at the time, it did not have a marketing information system, as such.

Whilst the sample of firms here is tiny it seems to support the findings of the Martech survey that, by and large, computerized marketing information systems in the UK are still in the early stages of development.

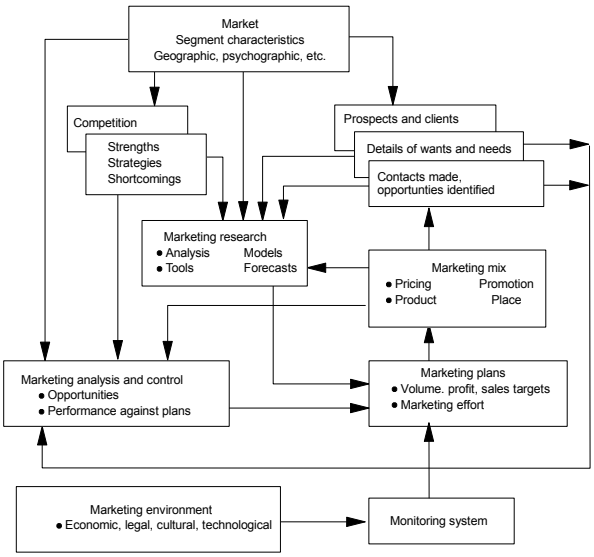
A Schema for Marketing Information Systems

Figure 1 shows a schema for an integrated marketing information system. The arrows indicate the flow of information between different components. Developing such an information system can take place piecemeal, and the system will not necessarily become integrated until all the components are in place and it is possible to pass information between the component parts with the minimum of effort.

Central to a marketing information system is the concept a database. There may be one or more databases present in such a system. The determination of this is a technical matter and it is difficult to state any hard and fast rules on this point. The various boxes in the diagram show the kind of information required by executives in the course of conducting marketing activities.

Information is required about the market and its various characteristics. Clearly, the kind of information held in the database should reflect how a firm segments its market. Indeed, many firms supply goods and services to more than one market so that the database will have to be organized in such a way that the individual markets can readily be identified. Different methods of segmenting the market may be employed for different products so that this too has to be taken into account. Quantitative information relating to the number of population members who have the various characteristics used as a basis for segmentation is also kept in the database. Not only does current quantitative information have to be held in the database but so also do past data. Without a substantial number of years' data it is difficult to forecast changes in the future.

Figure 1. Schema for a Marketing Information System



Closely related to the information held on the market itself is information held on competition and on prospects and clients. In the case of competition, it is important to have information on sales/market share and profitability of products by market segment. It is also useful to hold data on what competitors commit in the way of resources to products - classified by market segment. Armed with this information a firm can then analyze the data in the database and assess strengths and shortcomings of a competitor's product-market portfolio. Information about competitive strategy is more difficult to obtain. However, much can be learned from studying the quantitative data present in the database and, if historical data are kept, these can help to illuminate competitive strategy in an objective manner.

ONE NEEDS TO KEEP A COMPLETE LISTING OF ALL PROSPECTS AND ALL CLIENTS

In the case of prospects and clients one needs to keep a complete listing of all prospects and all clients, noting any purchases or purchase intentions they have. This information may be

obtained from both on-the-ground salespeople, salespeople in the order office and from any other person in the organization who is able to provide such information. Any special requirements or unfulfilled wants and needs should be recorded together with information regarding contacts made with actual or potential customers.

The marketing mix part of the information system database contains information on pricing, products, promotions and distribution strategies and policies both past and present together with current and previous evaluations of how different market segments respond to changes in these variables.

The marketing plans part of the information system database contains the current rolling marketing plan complete with the sales forecast and volume, profit and sales targets by products and market segment.

The data held in the marketing environment database relate to economic, legal, cultural and technological data. Much of the information will be qualitative and descriptive in nature and will tend to be used to aid judgment and decision making alongside quantitative analyses provided by the marketing analysis and control part of the information system.

The marketing analysis and control part of the marketing information system contains the "performance database" which provides information on what the firm has achieved along various dimensions. These include sales performance m unit terms, as well as in money terms; ROI and contribution to profit and overhead of individual products. They also include profitability of different market segments; information on achieved levels of distribution; effectiveness of advertising campaigns etc. The results of market analysis in identifying opportunities in the marketplace may also be placed here.

THERE IS ALWAYS A DANGER THAT TOO MUCH INFORMATION MAY BE ENTERED

The marketing research part of the marketing information system contains various models and other analytical tools. The analytical tools have access to data which are contained in any part of the database. Thus, it is possible to use these tools to compare marketing plans with what has in fact been achieved, or it is possible to analyze competitors' performance in relationship to the firm's performance.

The monitoring system looks at what is going on in the external environment. In particular it focuses on economic, legal, cultural and technological developments which have a bearing on the business. Such data need careful filtering to ensure that executives are not overloaded with information or presented with irrelevant information.

Tools of analysis

The surveys above underlined the absence of fully integrated marketing information systems. However; partial information systems or decision-support systems are needed when it comes to looking at marketing information handling and analysis. A database package and spreadsheets, together with free standing linear programming packages, elementary statistics packages, a forecasting tool and a project management tool (PERT) are likely to be the main tools of analysis. However, there are obviously many opportunities for developing and using a wide range of purpose-built decision-support aids.

Using a Marketing Information System

As we have seen above there are two basic. ingredients to a marketing information system. On the one hand, there is a database or a number of databases containing a variety of data about the firm, its competitors, its markets and the environment. On the other hand, there is the provision of a wide variety of analytical tools capable of exploring the data and turning it into meaningful information for management.

When designing a marketing information system a number of important questions need to be answered in the first place. These are:

(1) Exactly how much information will be entered in the database?
(2) What information will be entered into the database?
(3) How will it be entered into the database?
(4) How will it be manipulated once it is in the database?
(5) To whom will reports be sent?

The question of how much information is extremely important. There is always a danger that too much information may be entered. This will only serve to overload management's information

processing abilities. In addition, any data or information which is not used by management is clearly redundant and will be taking up valuable storage space in the information system. From time to time it is necessary to review the information available in the information system and to remove any that is not being used.

AN IMPORTANT SOURCE OF DATA IS THE INTERNAL ACCOUNTING SYSTEM

Closely allied to the question of how much information to enter is the question of what information should be entered. In order to be able to answer this question i~ ~ is important that the designer of the system carefully addresses the matter beforehand with people who will use the system. In addition, periodic reviews need to be f~' taken alter the system has been implemented to make sure that the system is still providing user satisfaction.

The question of how the data will be entered is linked to who will enter the data and what will be the sources of data. An important source of data is the internal accounting system of the company. This system reports orders, sales, stocks, debtors and creditors etc., and enables management to compare actual and expected levels of performance. It also enables management to spot opportunities and problems. The sales order/ invoicing system is the kernel of such a system. Often this is computerized and information can be made directly available to the marketing information system without any difficulty.

Competitive information and information on customers' wants and needs can be gleaned from salespeople's reports. Reports, of course, need to be filed into the information system in electronic form. This requires that either the sales staff have to do this themselves or, alternatively, someone else has to scan all sales reports and abstract information to put into the computerised system. The former method is decidedly more attractive but it calls for the design of electronic forms which can easily be completed by sales staff. Sales reports tend to be filled in at home at weekends or in hotel bedrooms during the working week. A portable laptop computer directly connected via a modem link with the firm's mainframe computer provides an attractive solution to the problem. Alternatively, floppy disk files handed in weekly to the computer support staff in the company could provide an adequate method

DATA SHOULD BE KEPT IN A DISAGGREGATED FORM IN THE DATABASE

Data on the environment and competitive activity can often bought from consultants and marketing research agencies in an electronic form.

Data should as far as possible be kept in a disaggregated form in the database. This allows anyone to manipulate and analyse the data to suit their own particular purposes. Summary statistical analyses of data may well be kept in a separate file within the database, if it is felt that it is information which people may want frequently.

Having a computer-based information system means that information in the form of reports can be made available quickly to management.

Sales management requires information to help it allocate the salesforce effectively and assess the performance of sales staff equitably. Sales staff, too, should be able to access the system easily and get support and information about such things as:

- The quantity of the product on hand.
- Prices and price discounts.
- Status information on invoices, time of delivery and back orders.
- Delivery dates.
- Complete product specifications.

The system should also aid the process of entering orders and reduce the salesperson's paperwork (see above: using a lap-top portable computer).

For control purposes sales performance analysis is required. This amounts to a detailed study of the total sales revenue of a company over a specific period of time. An analysis is made of total sales volume by product line, by salesperson, by territory and by customer groups. These sales are then compared with company goals and industry sales.

WE CAN EXPECT TO SEE CONSIDERABLE

DEVELOPMENTS IN THE NEXT FEW YEARS

Forecasting is the estimation of the market size and the company's share of that market. Marketing budgets, sales strategies and sales quotas are influenced by these estimates. The forecasts also help the planning and control of manufacturing, distribution management and advertising and promotion as they are reflected through the budgeting process. Information on sales profitability is also made available. This shows the relative profitability of customers, territories, product lines, etc.

Marketing research/intelligence helps to define marketing problems. It also helps executives find new customers and to adapt products to meet changing customer requirements. User-friendly software and large relational databases help advise users on which segments to target. Marketing research can tell management how to price a product, which distribution channels to use and how to get more out of advertising and other promotional expenditure.

Outputs of the marketing information system can take the form of reports. Example reports might be:

- A new product report, comprising an estimation of sales potential and customer buying habits and motives.
- A pricing strategy report to help management reach pricing objectives.
- A product-mix report to advise management on how to manage the product mix to best advantage - e.g. by changing the number of lines or the depth within a line or simply pruning or simplifying lines.
- A product life cycle report to help marketers manage the product through its various stages in the life cycle and possibly anticipate marketing requirements at a subsequent stage.
- An advertising effectiveness report to help assess who is the target audience, what to communicate, when to communicate and what media to use.
- A customer analysis report to spotlight customer trends, complaints and requests and a complete breakdown of profitability by customer.
- An order-processing control report to allocate stock to fill customer orders, process back orders, answer order status enquiries, product shipping reports by invoice and produce freight and labour costs.

Reports may be produced on a regular basis, as defined by the users of the system.

Conclusion

Computerised marketing information systems in many UK firms are largely in their infancy. However, with rapidly growing usage of mainframe, minicomputers and microcomputers we can expect to see considerable developments in this area in the next few years. There is a good deal to be done by many companies to improve their marketing information systems. To achieve this firms need help and assistance in choosing a system which best meets their own particular need. Obtaining a system is not by itself the solution to the problem, however. Employees need to be trained both how to use the system and how to operate it.

Many universities, polytechnics and colleges are now running information technology modules as part of the curricula in business and management courses. This underlines the growing importance of the subject area. Much more attention in business, however; has to be given to the need to implement information systems in the area of marketing.

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Source *Management Decision*, Vol. 29, No. 4,1991. pp. 55-60. ©MCB University Press Limited. 0025-1747